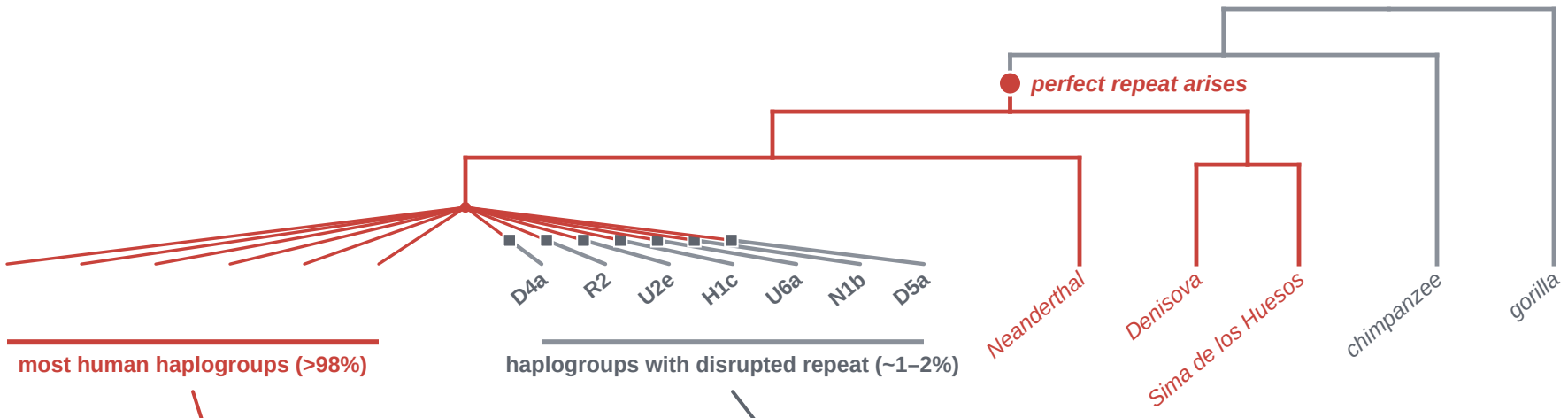
A. **The perfect common repeat arose in the *Homo* lineage and was disrupted again in a few human haplogroups**

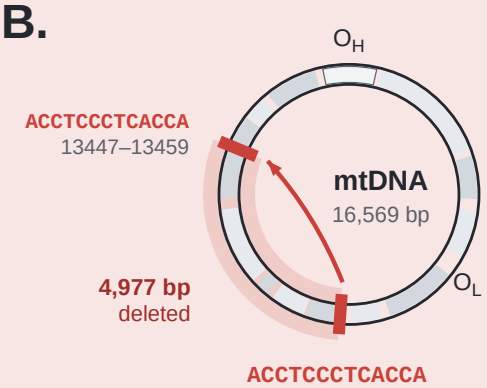


proximal arm, m.8470–8482

ACCTCCCTCACCA	**perfect**
ACC**C**CCCTCACCA	m.8473T>C D4a, R2, U2e, H1c, U6a
AC**T**TCCCTCACCA	m.8472C>T N1b
ACCTCCCTC**G**CCA	m.8479A>G D5a

PERFECT common repeat

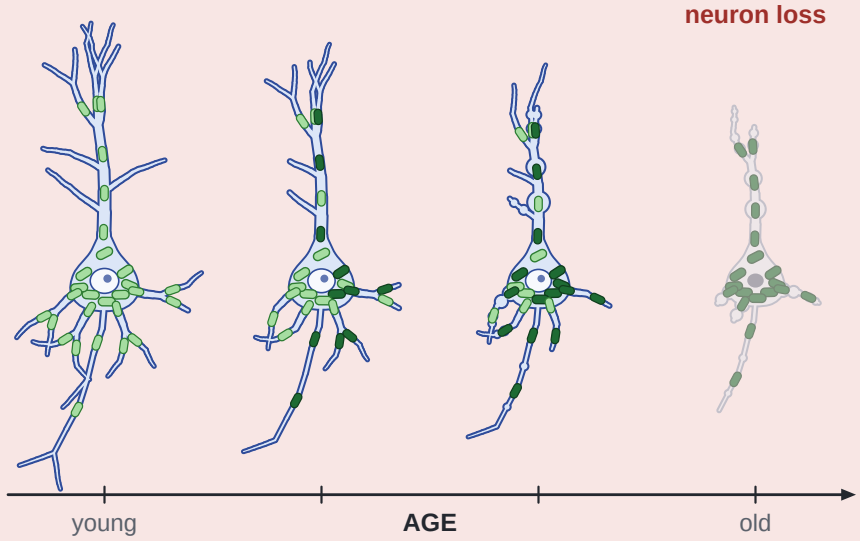
B.



perfect repeat:
13 of 13 bp identical

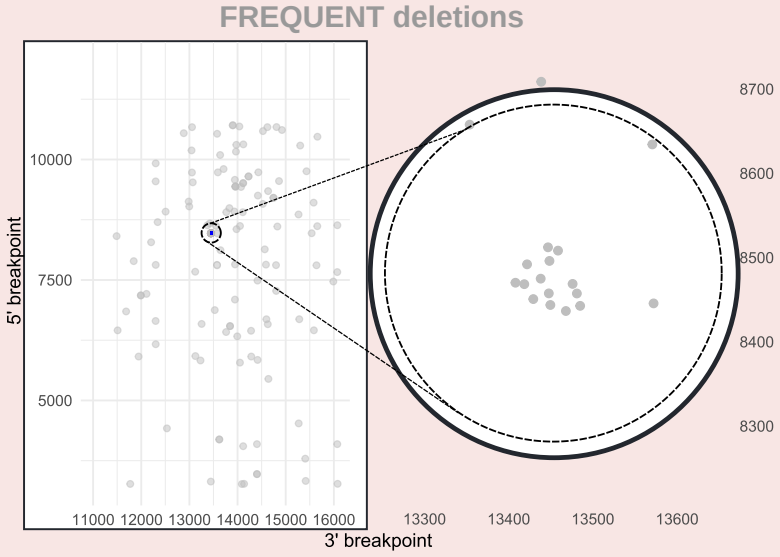
⇓

HIGH probability
of the common deletion

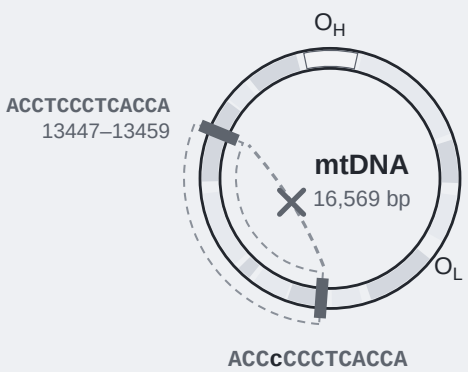


deletions clonally expand → accelerated ageing

FREQUENT deletions



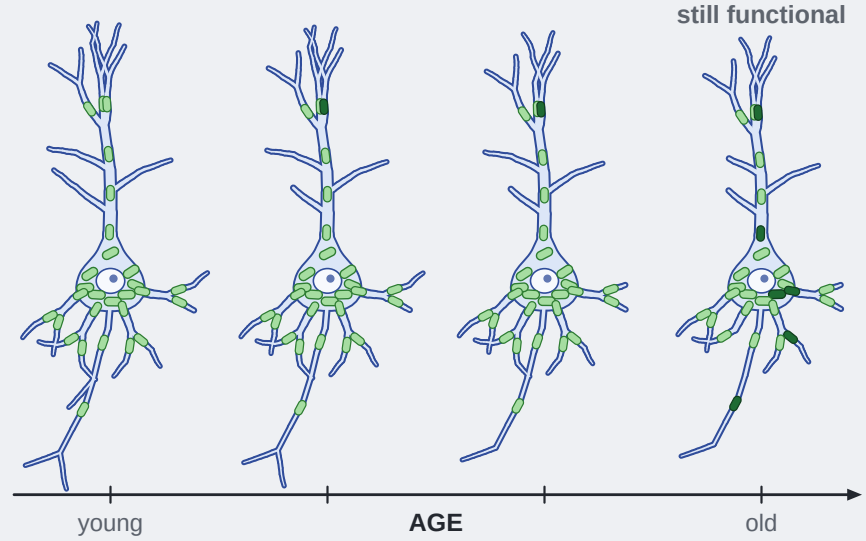
DISRUPTED common repeat



disrupted repeat:
one mismatch (12 of 13 bp)

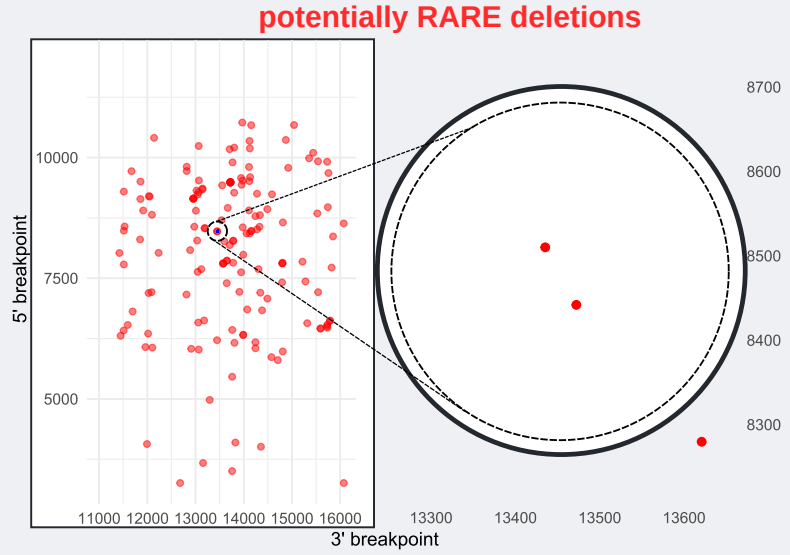
⇓

LOW probability
of the common deletion



few deletions → slower ageing

potentially RARE deletions



Data source: Guo et al., 2010

perfect common repeat | disrupted common repeat | mitochondrion, wild-type mtDNA | with deleted mtDNA (ΔmtDNA)